Federated Learning in Multi-Agent Systems: A Short Survey

Challenges, Opportunities, and the SkyData Setting

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1. Scope

This focused survey supports a research project whose theme is federated learning (FL) in multi-agent systems (MAS), with SkyData as the experimental ground. We review FL (Section 2), MAS (Section 3), their intersection including fully decentralised alternatives (Section 4), and locate the SkyData gap our companion paper addresses (Section 5).

2. Federated Learning

FL trains a shared model across clients that keep their data local, exchanging model updates rather than raw data; FedAvg alternates local training with periodic server-side averaging [4]. Surveys [8,15] organise its challenges into four themes, all relevant to MAS:

Challenge	Meaning	MAS relevance
Statistical heterogeneity	Non-IID client data	Agents live in different local environments
System heterogeneity	Unreliable devices/links	Agents join, leave, fail
Communication efficiency	Costly update exchange	Bandwidth and latency bound coordination
Fairness & robustness	Average can hurt atypical clients	Selfish or faulty agents must be tolerated

Two refinements matter for us. **Decentralised FL** removes the central server, aggregating over a peer topology [2]. **Agnostic FL** replaces the average objective by a min-max over client mixtures, protecting the worst-off participant [3].

3. Multi-Agent Systems

A MAS is a collection of autonomous agents perceiving, deciding and acting in a shared environment [5], characterised by autonomy, reactivity, pro-activeness and social ability [7]; deliberative agents often follow the Belief–Desire–Intention architecture [6]. Under partial observability, the central theme is coordination: good global behaviour from local decisions without a central controller — as in distributed coverage control [10].

4. The Intersection — and Its Fully Decentralised Frontier

FL and MAS meet when autonomous agents must learn collaboratively while preserving autonomy: non-IID and temporally correlated data, convergence under asynchronous or partially connected networks, tolerance of selfish agents, and the balance between local autonomy and global coordination. Three lines of work frame this frontier.

Gossip learning. Fully decentralised learning can dispense with any aggregator: nodes exchange and average models peer-to-peer along a communication graph [11,12]. Convergence then depends on the connectivity and mixing properties of that graph — peers must be able to reach one another, directly or by diffusion. This is the closest conceptual relative of aggregation in SkyData, and the sharpest contrast: SkyData guarantees no path between two agents at all [1], so no overlay can be assumed.

Partial client participation. A substantial FL literature analyses convergence when only a fraction of clients participates per round — through sampling strategies [14] or arbitrary availability patterns [13]. There, participation is a design or scheduling matter over a fixed client pool. In autonomous-data settings, non-participation is a property of the environment: participants are mobile, and none is guaranteed to return.

Federated estimation. Between full model training and no learning lies collaborative estimation of environment parameters: agents share estimates — not observations — of quantities they each see partially. Small models make the mechanism inspectable and the benefit of sharing measurable; this is the form of FL our companion paper instantiates.

5. The SkyData Gap

SkyData [1] gives data autonomy: each replica is an agent (SKD); replicas of one datum form a family; dataless SkyWorkers perform system tasks — gathering and aggregating models is their archetype — then vanish, holding no authority [1]. The conjunction that no surveyed line handles together: no permanent authority (unlike replica-placement services [9]), no guaranteed reachability (unlike gossip overlays and FL client pools), full agent autonomy. Our companion paper addresses one well-scoped piece: a SkyWorker that both coordinates a family's migrations (sequential reservation) and federates its knowledge (weighted averaging of harbour-quality estimates), with the honest finding that coordination provides stability whose value grows with the agents' view, and federation roughly halves collective discovery time under partial observability.

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